Math Notes

August 2025

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1 Matrix Theory

1.1 Matrix Decompositions and Inertia

1. CS Decomposition for $X = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} \in \mathrm{U}(n)$ with $X_{11} \in \mathbb{C}^{k\times k}$, $2k \le n$. (Carefully adopted from [4, Theorem 2.5.3], or [8, Section 4])

$$X = \operatorname{diag}(U_1, U_2) \begin{bmatrix} C & -S & 0 \\ S & C & 0 \\ 0 & 0 & I_{n-2k} \end{bmatrix} \operatorname{diag}(V_1, V_2)^*$$

where $U_1, V_1 \in \mathrm{U}(k), U_2, V_2 \in \mathrm{U}(n-k), C = \operatorname{diag}(\cos\theta_1, \dots, \cos\theta_k), S = \operatorname{diag}(\sin\theta_1, \dots, \sin\theta_k), \theta_i \in [0, \pi/2]$.

2. (Haynsworth inertia additivity formula) Let $H = \begin{bmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{bmatrix} \in \mathbb{C}^{n\times n}$ be hermitian and its inertia be a triple $\mathrm{In}(H) \coloneqq (\pi(H), \nu(H), \delta(H))$, whose components are the numbers of positive, negative and zero eigenvalues of H. Assume H_{11} is nonsingular. Then $\mathrm{In}(H) = \mathrm{In}(H_{11}) + \mathrm{In}(H/H_{11})$, where $H/H_{11} = H_{22} - H_{12}^* H_{11}^{-1} H_{12}$ is the Schur complement.

1.2 Quaternions and Their Representations

1. $q = a + b\mathrm{i} + c\mathrm{j} + d\mathrm{k} \in \mathbb{H}$, where $\mathrm{ij} = \mathrm{k}, \mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = -1$.
2. Denote $q = q_1 + q_2\mathrm{j}$ with $q_1, q_2 \in \mathbb{C}$. Denote $\mathfrak{R}(q) \coloneqq a$, $\mathfrak{V}(q) \coloneqq q - \mathfrak{R}(q)$.
3. Quaternion conjugate $\overline{q} \coloneqq a - b\mathrm{i} - c\mathrm{j} - d\mathrm{k} = \overline{q_1} - q_2\mathrm{j}$.
4. Representation via complex or real numbers. Let $\rho_1 : \mathbb{H} \to \mathbb{C}^{2\times 2}$, $\rho_2 : \mathbb{C} \to \mathbb{R}^{2\times 2}$, $\rho_3 : \mathbb{H} \to \mathbb{R}^{4\times 4}$ be defined as

$$\rho_1 \coloneqq \begin{bmatrix} q_1 & q_2 \\ -\overline{q_2} & \overline{q_1} \end{bmatrix}, \text{ where } q_1 = a + b\mathrm{i},\ q_2 = c + d\mathrm{i},$$

$$\rho_2(a + b\mathrm{i}) \coloneqq \begin{bmatrix} a & -b \\ b & a \end{bmatrix},$$

$$\rho_3(q) \coloneqq \rho_2 \circ \rho_1(q) = \begin{bmatrix} a & -b & c & -d \\ b & a & d & c \\ -c & -d & a & b \\ d & -c & -b & a \end{bmatrix}.$$

5. For $q, q^{(i)} \in \mathbb{H}$,

$$\rho_1(q^{(1)}q^{(2)}) = \rho_1(q^{(1)})\rho_1(q^{(2)}), \quad \rho_1(\overline{q}) = \rho_1(q)^{\mathsf{H}};$$
$$\rho_3(q^{(1)}q^{(2)}) = \rho_3(q^{(1)})\rho_3(q^{(2)}), \quad \rho_3(\overline{q}) = \rho_3(q)^{\mathsf{T}}.$$

1.3 Quaternionic Matrices

1. For $A \in \mathbb{H}^{n\times n}$, $A = A_1 + A_2\mathrm{j}$ with $A_1, A_2 \in \mathbb{C}^{n\times n}$.

   Complex representation:

$$\psi(A) \coloneqq \begin{bmatrix} A_1 & A_2 \\ -\overline{A_2} & \overline{A_1} \end{bmatrix}$$

$$A^* \coloneqq A_1^{\mathsf{H}} - A_2^{\mathsf{T}}\mathrm{j}$$

   where the operation A_1^{H} on $A_1 \in \mathbb{C}^{n\times n}$ just means transpose and complex conjugate.

$$AB = (A_1B_1 - A_2\overline{B_2}) + (A_1B_2 + A_2\overline{B_1})\mathrm{j}$$

$$\psi(AB) = \psi(A)\psi(B),\ \psi(A^*) = \psi(A)^{\mathsf{H}}.$$

2. Define $\mathrm{U}(n,\mathbb{H}) \coloneqq \{X \in \mathbb{H}^{n\times n} :\ XX^{\mathsf{H}} = I\}$. $\mathrm{Sp}(n) = \mathrm{Sp}(2n,\mathbb{C}) \cap \mathrm{U}(2n)$. Then ψ is a group isomorphism between them.

3. Real representation: now change the notation to $A = A_1 + A_2\mathrm{i} + A_3\mathrm{j} + A_4\mathrm{k}$ with $A_1, \ldots, A_4 \in \mathbb{R}^{n\times n}$, then

$$\varphi(A) \coloneqq \begin{bmatrix} A_1 & -A_2 & A_3 & -A_4 \\ A_2 & A_1 & A_4 & A_3 \\ -A_3 & -A_4 & A_1 & A_2 \\ A_4 & -A_3 & -A_2 & A_1 \end{bmatrix}$$

$$\varphi(AB) = \varphi(A)\varphi(B),\ \varphi(A^*) = \varphi(A)^{\mathsf{T}}.$$

4. (Corollary) For $\alpha, \beta \in \mathbb{H}$, $A, B \in \mathbb{H}^{m\times n}$, [9, Definition 3.2.1]

$$(\alpha A + \beta B)^* = A^*\alpha^* + B^*\beta^*,\ (AB)^* = B^*A^*,\ (A^*)^* = A.$$

1.3.1 Spectrum, EVD, and SVD

1. [9, Chapter 2.2] Similarity. $\mathrm{Sim}(x) \coloneqq \{y \in \mathbb{H} :\ axa^{-1} = y \text{ for some } a \in \mathbb{H}\backslash\{0\}\}$.

2. Congruence. $\mathrm{Con}(x) \coloneqq \{y \in \mathbb{H} :\ axa^* = y \text{ for some } a \in \mathbb{H}\backslash\{0\}\} = \bigcup_{\lambda>0}\{\lambda\mathrm{Sim}(x)\}$.

3. $\mathrm{Sim}(x) = \mathfrak{R}(x) + |\mathfrak{V}(x)|\mathbb{S}$, where the unit sphere $\mathbb{S} \coloneqq \{q \in \mathbb{H} :\ \mathfrak{R}(q) = 0,\ |q| = 1\}$.

4. For $A \in \mathbb{H}^{n\times n}$, $(v,\alpha) \in (\mathbb{H}^n\backslash\{0\}) \times \mathbb{H}$ is a right eigenvector/eigenvalue of A if $Av = v\alpha$. Denote the set of all right eigenvalues as the spectrum $\sigma(A)$.

5. If $\alpha \in \sigma(A)$, then $\mathrm{Sim}(\alpha) \subseteq \sigma(A)$. ($A(v\lambda) = (v\lambda)(\lambda^{-1}\alpha\lambda)$, $\lambda \neq 0$.)

6. [9, Theorem 5.3.6] If $A \in \mathrm{U}(n, \mathbb{H})$, then there exists $U \in \mathrm{U}(n, \mathbb{H})$, $D = \mathrm{diag}(e^{\mathrm{i}\theta_1}, \ldots, e^{\mathrm{i}\theta_n})$ with $\theta_i \in [0, \pi]$, s.t. $A = UDU^*$.

7. If $A \in \mathbb{H}^{n\times n}$ is hermitian ($A = A^*$), there exists $U \in \mathrm{U}(n, \mathbb{H})$ s.t. U^*AU is real and diagonal.

8. [9, Proposition 3.2.5] For $A \in \mathbb{H}^{m\times n}$, there exists SVD

$$A = U \begin{bmatrix} \mathrm{diag}(a_1, \ldots, a_k) & 0 \\ 0 & 0_{(m-k)\times(n-k)} \end{bmatrix} V^*$$

with unique singular values $a_1 \geq \cdots \geq a_k > 0$.

9. Because $\mathbb{H}$ is not commutative, we define determinant and trace through complex representations. For $A \in \mathbb{H}^{n\times n}$, $\mathrm{tr}_{\mathbb{C}}(A) := 2\mathfrak{R}(\sum_{i=1}^{n} a_{ii}) = \mathrm{tr}(\psi(A))$. [9, Section 5.9] [2, Definition 2.1]

1.4 Symplectic and Hamiltonian Matrices

1. Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$.

2. $A \in \mathbb{C}^{2n\times 2n}$ is called Hamiltonian if $J^{-1}A^{\intercal}J = -A$ (equivalently, $(JA)^{\intercal} = JA$). Then $A_1 = -A_4^{\intercal}$, and A_2, A_3 are symmetric.

3. Skew-Hamiltonian if $J^{-1}A^{\intercal}J = A$. Then $A_1 = A_4^{\intercal}$, and A_2, A_3 are skew-symmetric.

4. [3, Proposition 1] $A \in \mathrm{Sp}(2n, \mathbb{C})$ if and only if $A^{\intercal}JA = J$, or $JA^{-1} = A^{\intercal}J$. Then $A_{11}A_{22}^{\intercal} - A_{12}A_{21}^{\intercal} = I$, $A_{11}A_{12}^{\intercal}$ and $A_{21}A_{22}^{\intercal}$ symmetric.

5. If $A \in \mathrm{Sp}(2n, \mathbb{C})$, so is $A^{-1}, A^{\intercal}, \overline{A}, A^*$.

6. [3, Lemma 3] (double-cited) Let A, B be symplectic. Then A, B are similar if and only if they are symplectically similar.

7. Denote expanding sum: $A \boxplus B = \begin{bmatrix} A_{11} \oplus B_{11} & \ldots \\ \ldots & A_{22} \oplus B_{22} \end{bmatrix}$.

8. [3, Lemma 5] Each symplectic matrix is symplectically similar to the expanding sum of matrices of the form:

$$J_k(\lambda) \oplus J_k(\lambda)^{-\intercal} \quad (\lambda \neq 0, \pm 1),$$
$$J_{2k-1}(\varepsilon) \oplus J_{2k-1}(\varepsilon)^{-\intercal} \quad (\varepsilon = \pm 1),$$
$$\pm F_{2k} = \pm \begin{bmatrix} J_k(1) & \begin{bmatrix} 0_{k-1,k} \\ u \end{bmatrix} \\ 0 & J_k(1)^{-\intercal} \end{bmatrix}$$

where $u = [(-1)^{k-1}, (-1)^{k-2}, \ldots, 1]$.

9. (Williamson's Thm) Let $M \in \mathbb{R}^{2n\times 2n}$ be a strictly positive definite real symmetric matrix. Then there exists $S \in \mathrm{Sp}(n,\mathbb{R})$ and a positive diagonal D such that $SMS^{\mathsf{T}} = \mathrm{diag}(D,D)$. For generalization to positive semi-definite matrices and real symmetric matrices, see [7].

1.5 Compact Symplectic Groups

1. (Compact symplectic group) As $\mathbb{C}^{2n\times 2n}$ block matrices:

$$\mathrm{Sp}(n) = \left\{ \begin{bmatrix} A & B \\ -\overline{B} & \overline{A} \end{bmatrix} : \begin{array}{c} A,B \in \mathbb{C}^{n\times n}, \\ AA^{\mathsf{H}} + BB^{\mathsf{H}} = I_n, \quad AB^{\mathsf{T}} = BA^{\mathsf{T}} \end{array} \right\},$$

$$\mathrm{Sp}(n) = \left\{ Q \begin{bmatrix} D & \\ & \overline{D} \end{bmatrix} Q^{\mathsf{H}} : \begin{array}{c} Q \in \mathrm{Sp}(n), \\ D = \mathrm{diag}(e^{\mathrm{i}\theta_1}, \ldots, e^{\mathrm{i}\theta_n}), \\ \theta_i \in [0, 2\pi) \end{array} \right\}.$$

$$\mathfrak{sp}(n) \coloneqq \left\{ \begin{bmatrix} A & B \\ -\overline{B} & \overline{A} \end{bmatrix} : A = -A^{\mathsf{H}},\ B = B^{\mathsf{T}} \right\}$$

1.5.1 Unitary Groups as Real Symplectic Matrices

1. ($\mathrm{U}(n) \cong \mathrm{SO}(2n) \cap \mathrm{Sp}(2n,\mathbb{R})$) Let $A = B + \mathrm{i}C \in \mathrm{U}(n)$. Let $\rho : \mathrm{U}(n) \to \mathrm{SO}(2n)$, $\rho(A) \coloneqq \begin{bmatrix} B & -C \\ C & B \end{bmatrix} = P\,\mathrm{diag}(A,\overline{A})P^{-1}$, where $P = \begin{bmatrix} I & I \\ -\mathrm{i}I & \mathrm{i}I \end{bmatrix}$. ρ is a group isomorphism.

For tangent spaces and matrix orbits, see Section 4.2.

2 Algebras

2.1 Algebra Basics

1. (Algebra) An algebra A over field K is a vector space with an additional product operation $\cdot : A \times A \to A$, such that the following holds for all $x,y,z \in A$, $a,b \in K$:
   (a) Right distributivity: $(x+y)\cdot z = x\cdot z + y\cdot z$
   (b) Left distributivity: $z\cdot(x+y) = z\cdot x + z\cdot y$
   (c) Compatibility with scalars: $(ax)\cdot(by) = (ab)(x\cdot y)$
2. (Unital, Zero)

2.2 Clifford Algebras

1. Let V be a vector space, K be a field. Let $Q : V \to K$ be a quadratic form.

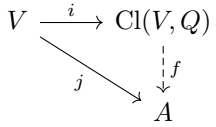
2. Tensor algebra $T(V) \coloneqq \bigoplus_{d=0}^{\infty} V^{\otimes d}$. The two-sided ideal

$$I_Q = \langle v^2 - Q(v)1 \rangle \coloneqq \{a(v^2 - Q(v)1)b : a, b \in T(V)\}.$$

   The Clifford algebra is $\mathrm{Cl}(V, Q) \coloneqq T(V)/I_Q$.

3. K-bilinear form associated with Q is $B(u, v) \coloneqq \frac{1}{2}(Q(u+v) - Q(u) - Q(v))$. So $uv + vu = 2B(u, v)1$ [1], and $Q(v) = B(v, v)$.

4. (Abstract definition) A Clifford algebra $\mathrm{Cl}(V, Q)$ is a pair (B, i), where B is a unital associative algebra of K and i is a linear map $i : V \to B$ such that $i(v)^2 = Q(v)1_B$.
   (Universal property) Given any unital associative algebra A over K and linear map $j : V \to A$ such that $j(v)^2 = Q(v)1_A$, there is a unique algebra homomorphism $f : B \to A$ such that $f \circ i = j$. The commutative diagram:



5. (Classification of Clifford algebra with $K = \mathbb{R}$ or $\mathbb{C}$)

6. `https://math.stackexchange.com/questions/4925361/clifford-algebra-for-hermitian-forms`
   `https://math.stackexchange.com/questions/4653580/are-there-unitary-clifford-alegbras-in-analogy-with-orthogonal-and-symplectic/4653718#4653718`

2.3 CAR Algebras

2.4 Weyl Algebras

2.5 $*$-Algebras and C^*-Algebras

1. (*-ring) R is a ring with an antiautomorphism/involution map $* : R \to R$ such that for all $x, y \in R$,

   (a) $(x + y)^* = x^* + y^*$, $(xy)^* = y^* x^*$

   (b) $1^* = 1$, $(x^*)^* = x$

[1] Assume 2 is invertible in K.

2. (*-algebra) A is a *-algebra, if A is an associative algebra and is a *-ring with involution $*$ over a commutative *-ring R with involution $'$, such that for all $r \in R, x \in A$, $(rx)^* = r'x^*$.
   Since A is *-ring, we have for $x, y \in A$, $\lambda, \mu \in R$
   (a) $x^{**} = x$, $(xy)^* = y^*x^*$
   (b) $(\lambda x + \mu y)^* = \lambda' x^* + \mu' y^*$

3. (*-homomorphism) an algebra homomorphism $f : A \to B$ that is also compatible with the involutions of A, B, i.e., $f(a^*) = f(a)^*$ for all $a \in A$.
   (Example) $B = \mathbb{R}^{2n\times 2n}$ with $*$ being transpose over base ring $R_B = \mathbb{R}$ with trivial identical involution $r' = r$. $A = \mathbb{C}^{n\times n}$ with $*$ being conjugate transpose over base ring $R_A = \mathbb{C}$. The embedding f of $\mathbb{C}^{n\times n}$ into $\mathbb{R}^{2n\times 2n}$ is a *-homomorphism.

4. (Banach algebra) is an associative algebra A over $\mathbb{R}$ or $\mathbb{C}$ and is a Banach space with norm satisfying $\|xy\| \leq \|x\|\|y\|$ fo all $x, y \in A$.

5. (C*-algebra) A is a Banach and *-algebra over $\mathbb{C}$, with the C* identity $\|xx^*\| = \|x\|^2$ for all $x \in A$.
   (Example) $A = \mathbb{C}^{n\times n}$ with operator norm $\|\cdot\|_2$.

3 Topology and Differential Geometry

3.1 Topological Spaces and Covers

1. Let (X, τ) be a topological space. τ is a family of subsets of X that satisfies
   (a) $\varnothing \in \tau$, $X \in \tau$.
   (b) If $S_\alpha \in \tau$, then $\bigcup_{\alpha\in A} S_\alpha \in \tau$, where A may be any index set (finite, countable, or uncountable).
   (c) If $S_i \in \tau$ and $\mathbb{N} \ni n < \infty$, then $\bigcap_{i=1}^{n} S_i \in \tau$.

   If $S \in \tau$, then S is called an open set.

2. (Subspace topology) Given $S \subseteq X$, define $\tau_S \coloneqq \{S \cap U : U \in \tau\}$.

3. (Cover) The collection $\{U_\alpha : \alpha \in A\}$ is a cover of X if $\bigcup\limits_{\alpha\in A} U_\alpha = X$.

4. (Refinement) Let $C = \{U_\alpha\}_{\alpha\in A}, D = \{V_\beta\}_{\beta\in B}$ be two covers of X. D is a refinement of C if for all $\beta \in B$ there exists $\alpha \in A$ such that $V_\beta \subseteq U_\alpha$.

5. (Locally finite) A collection of subsets $\{U_\alpha\}_{\alpha\in A}$ is locally finite if each $x \in X$ has a neighborhood N_x that only intersects finitely many sets in the collection.

6. (Paracompact)

3.2 Connectedness, Homotopy, and Fundamental Groups

1. (Connected, path-connected, simply connected) X is connected if it cannot be divided into two disjoint non-empty open sets. X is path-connected if every $x, y \in X$ can be connected by a path. X is simply connected if it is path-connected and any two paths with same endpoints are homotopic to each other.

   (a) connected $\Rightarrow$ path-connected $\Rightarrow$ simply connected

   (b) Connected but not path-connected: the topologist's sine curve $T \coloneqq \{(x, \sin(1/x)) : x \in (0, 1]\} \cup \{(0, 0)\}$ under the topology of the Euclidean plane.

   (c)

2. (Homotopy) Let $f, g : X \to Y$ be continuous. A homotopy between f and g is a continuous function $H : X \times [0, 1] \to Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$.

3. (fundamental group) A loop based at x_0 is a continuous function $\gamma : [0, 1] \to X$ such that $\gamma(0) = \gamma(1) = x_0$. The fundamental group of the topological space X is

$$\pi_1(X, x_0) \coloneqq \{\text{all loops } \gamma \text{ based at } x_0\}/\text{homotopy}.$$

   (a) Given two loops γ_0, γ_1, $(\gamma_0 \cdot \gamma_1)(t) = \begin{cases} \gamma_0(2t) & 0 \le t \le \frac{1}{2}, \\ \gamma_1(2t-1) & \frac{1}{2} < t \le 1. \end{cases}$ (in the sense of equivalent classes, $[\gamma_0] \cdot [\gamma_1] = [\gamma_0 \cdot \gamma_1]$). $\gamma^{-1}(t) = \gamma(1-t)$. The identity element is the constant loop $e(t) = x_0$.

   (b) If X is path-connected, $\pi_1(X, x_0)$ doesn't depend on x_0.

3.3 Local Homeomorphisms and Covering Spaces

1. (Local homeomorphism) A function $f : X \to Y$ between two topological spaces is called a local homeomorphism if every $x \in X$ has a open neighborhood U, such that $f(U)$ is open in Y and $f|_U : U \to f(U)$ is a homeomorphism.

2. Topological spaces X, Y have same homotopy type if there exist $f : X \to Y$ and $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_Y$, where $\simeq$ stands for homotopy equivalence. Homeomorphism $\Rightarrow$ of same homotopy type.

3. (Covering projection) A covering of X is a continuous map

$$\pi : \tilde{X} \to X$$

   such that for every $x \in X$ there exists an open neighborhood U_x of x and a discrete space D_x such that $\pi^{-1}(U_x)$ is the disjoint union $\bigcup_{d \in D_x} V_d$ and

$$\pi|_{V_d} : V_d \to U_x$$

is a homeomorphism for every $d \in D_x$. The open sets V_d are called sheets. $\pi^{-1}(x)$ is called the fiber of x.

3.4 Manifolds and Tangent Spaces

1. A chart for a topological space M is a homeomorphism φ from an open subset $U \subseteq M$ to an open subset of a Euclidean space.

2. An atlas for M is an indexed family $\{(U_\alpha, \varphi_\alpha) : \alpha \in I\}$ of charts on M s.t. $\bigcup_{\alpha \in I} U_\alpha = M$.

3. If each U_α is an open subset of n-dimensional Euclidean space, then M is said to be an n-dimensional manifold.

4. Transition maps. Let $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are two charts for M s.t. $U_\alpha \cap U_\beta \neq \varnothing$. The transition map $\tau_{\alpha,\beta} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is defined as $\tau_{\alpha,\beta} \coloneqq \varphi_\beta \circ \varphi_\alpha^{-1}$.

5. (Tangent space T_xM defined through equivalent classes of curves.) For $x \in M$ and a smooth curve $\alpha : I \to M$ with $I \subseteq \mathbb{R}$, $\alpha(0) = x$. Let (U, ϕ) be a chart of M with $x \in U$. Then $\phi \circ \alpha : I \to \phi(U) \subseteq \mathbb{R}^n$ is a smooth map and $(\phi\alpha)'(0)$ is its velocity at 0 w.r.t. the current chart. We say two curves through x are equivalent, i.e. $\alpha \sim_x \beta$, if $(\phi\alpha)'(0) = (\phi\beta)'(0)$ (Can prove this relation is chart-independent.) Then each equivalent class $[\alpha]_x$ is an element in T_xM. [6, Definition C.5.1]

   Given the chart (U, ϕ), $\tau_x^\phi : T_xM \to \mathbb{R}^n$, $\tau_x^\phi([\alpha]_x) = (\phi\alpha)'(0)$ is an isomorphism between two vector spaces.

6. (Submanifold) Let M be a n-dimensional smooth manifold and $A \subseteq M$. If every $a \in A$ possesses a chart (U, ϕ) s.t. $a \in U$ and $\phi(A \cap U) = \phi(U) \cap (\mathbb{R}^k \times \{0\})$ for some k, then A is a smooth submanifold of M.

7. (Pullback) Let $f : M \to N$ be smooth, and let ω be a smooth k-form on N. Then there is a differentialn form $f^*\omega$ on M called the pullback of ω. For $p \in M$ and $v_1, \ldots, v_k \in T_pM$,

$$(f^*\omega)_p(v_1, \ldots, v_k) = \omega_{f(p)}(df_p(v_1), \ldots, df_p(v_k)).$$

8. (Immersion, Embedding, Submersion) If $T_pf : T_pM \to T_{f(p)}N$ is injective for all $p \in M$, then f is called an immersion. If f is an immersion and $f : M \to f(M) \subseteq N$ is a homeomorphism, then f is called an embedding. If T_pf is surjective for all $p \in M$, then f is called a submersion.

3.5 Differentials, Critical Values, and Rank Theorems

1. Let M, N be differentiable manifolds of dimension m and n respectively. Let $f : M \to N$ be smooth. The differential is $Df : TM \to TN$.

2. (Sard's Thm) Let $X \coloneqq \{x \in N : \text{ rank}(Df_x) < n\}$ be the set of critical points. Then the set of critical values $f(X)$ has measure zero in N.

3. (Constant rank Thm) If $\text{rank}(Df) \equiv r$ in a neighborhood of $x \in M$, then locally there exist coordinate charts φ, ψ such that $(\psi \circ f \circ \varphi)(u_1, \ldots, u_m) = (u_1, \ldots, u_r, \underbrace{0, \ldots, 0}_{n-r})$.

4. (Corollary) If $\text{rank}(Df_x) < n$ for all $x \in M$, then f is non-surjective.

3.6 Affine and Levi-Civita Connections

Stick to [1].

1. (Affine connection, [1, Section 5.2]) Let $\mathfrak{X}(M)$ denote the set of smooth vector fields on manifold M, $\mathfrak{F}(M)$ the set of smooth, real-valued functions on M.

   Let $f, g \in \mathfrak{F}(M)$ be functions and $\eta, \chi, \xi, \zeta \in \mathfrak{X}(M)$ be vector fields. An affine connection on M is a mapping

   $$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M), \quad (\eta, \xi) \overset{\nabla}{\to} \nabla_\eta \xi$$

   and satisfies

   (a) $\mathfrak{F}(M)$-linearity in η: $\nabla_{f\eta + g\chi}\xi = f\nabla_\eta \xi + g\nabla_\chi \xi$,
   (b) $\mathbb{R}$-linearity in ξ: $\nabla_\eta(a\xi + b\zeta) = a\nabla_\eta \xi + b\nabla_\eta \zeta$,
   (c) Product rule: $\nabla_\eta(f\xi) = (\eta f)\xi + f\nabla_\eta \xi$.

   Note that for $\eta, \xi \in \mathfrak{X}(M), f \in \mathfrak{F}(M)$, we have $\eta f \in \mathfrak{F}(M)$ (directional derivative of f along η), and $f\xi \in \mathfrak{X}(M)$ (scalar-vector product) [1, Section 3.5.4].

2. On an arbitrary (second-countable Hausdorff) manifold, there are infinitely many affine connections.

3. All possible affine connections of the Euclidean space $\mathbb{R}^n$ are characterized by n^3 real-valued functions Γ_{ij}^k on $\mathbb{R}^n$, a.k.a. Christoffel symbols.

4. (Levi-Civita connection, [1, Theorem 5.3.1]) On a Riemannian manifold M, there exists a unique affine connection ∇ that satisfies

   (a) $\nabla_\eta \xi - \nabla_\xi \eta = [\eta, \xi]$ (symmetry),
   (b) $\chi\langle \eta, \xi \rangle = \langle \nabla_\chi \eta, \xi \rangle + \langle \eta, \nabla_\chi \xi \rangle$ (compatibility with the Riemannian metric),

   for all $\chi, \eta, \xi \in \mathfrak{X}(M)$.

5. Curvature tensor, Sectional curvature

3.7 Morse Theory

1. (Morse Lemma) Let p be a non-degenerate critical point (Hessian non-singular) of $f : M \to \mathbb{R}$. Then there exists a chart $(x_1, \ldots, x_n)$ in a neighborhood U of p s.t. $x_i(p) = 0$ for all i and $f(x) = f(p) - \sum_{i=1}^{\gamma} x_i^2 + \sum_{i=\gamma+1}^{n} x_i^2$ on U. Here γ is the index of f at p.

2. Non-degenerate critical points are isolated.

3. $f : M \to \mathbb{R}$ is called a Morse function if it is smooth and has no degenerate critical points.

4. $f : M \to \mathbb{R}$ is called a Morse-Bott function if it is smooth and the set C of critical points is a closed submanifolds of M, such that for all $x \in C$, $T_x C = \ker(H_f(x))$. (The Hessian is non-singular when restricted to directions normal to the submanifold C.)

5. {Morse functions} $\subsetneq$ {Morse-Bott functions}. Isolated critical points are 0-dimensional submanifolds.

6. (Morse-Bott Lemma) Generalize the Morse Lemma.

$$f(x) = f(p) - \sum_{i=1}^{n^-} x_i^2 + \sum_{i=n^-+1}^{n^-+n^+} x_i^2$$

   on U, where $n^0 \coloneqq n - n^- - n^+ = \dim C$.

4 Lie Groups and Lie Algebras

4.1 Lie Groups and Group Actions

1. G is a Lie group if it is a group and a smooth manifold s.t. $G \times G \to G$, $(x, y) \mapsto xy^{-1}$ is a smooth map. The Lie algebra of G is $\mathfrak{g} \coloneqq T_e G$.

2. (Action) A Lie group action of G on a smooth manifold M is a smooth map

$$\sigma : G \times M \to M,\ (g, x) \mapsto g \cdot x$$

   satisfying $\forall g, h \in G, \forall x \in M,\ g \cdot (h \cdot x) = (gh) \cdot x, e \cdot x = x$. The group action σ is called transitive if for any $x, y \in M$ there exists $g \in G$ s.t. $y = g \cdot x$. M is called homogeneous if there exists a transitive G-action on M.

3. Each orbit of the group action is one equivalent class. The orbit $\mathcal{O}(x) \coloneqq \{g \cdot x : g \in G\}$. If G is a compact Lie group, $\mathcal{O}(x)$ is a smooth, compact submanifold of M. $M/G \coloneqq \{\mathcal{O}(x) : x \in M\}$. [6, Section 2.1]

4. The stabilizer $\mathrm{Stab}(x) \coloneqq \{g \in G :\ g \cdot x = x\}$. For a subgroup $H \leq G$, the set of coset classes $G/H \coloneqq \{g \cdot H : g \in G\}$. [6, Section 2.1]

5. For fixed $x \in M$, $\overline{\sigma}_x : G/\mathrm{Stab}(x) \to \mathcal{O}(x),\ g\mathrm{Stab}(x) \mapsto g \cdot x$ is a smooth bijection. (The orbit-stabilizer Thm naturally follows.) $\overline{\sigma}_x$ is a diffeomorphism if G is a compact Lie group. [6, Section 2.1]

6. Isometry group of a metric space V is the set of all bijective isometries (distance-preserving maps) from V to itself.

4.2 Matrix Groups and Orbit Geometry

For concrete quaternionic and symplectic matrix representations, see Section 1.

1. $T_Q\,\mathrm{O}(n) = \{QX \in \mathbb{R}^{n\times n} :\ X + X^\intercal = 0\} = \{XQ \in \mathbb{R}^{n\times n} :\ X + X^\intercal = 0\}$. Since $XQ = Q\underbrace{Q^\intercal XQ}_{X'}$. Similar structure holds for $\mathrm{U}(n), \mathrm{U}(n,\mathbb{H})$.

2. $T_Q\,\mathrm{U}(n,\mathbb{H}) = \{QX \in \mathbb{H}^{n\times n} :\ X + X^* = 0\}$.

3. $\dim_{\mathbb{R}} \mathrm{U}(n,\mathbb{H}) = 2n^2 + n = \dim_{\mathbb{R}}\{X \in \mathbb{H}^{n\times n} :\ X + X^* = 0\}$.

4. Let $Q = \mathrm{diag}(\lambda_1 I_{n_1}, \ldots, \lambda_r I_{n_r})$, $M(Q) \coloneqq \{OQO^\intercal : O \in \mathrm{O}(n)\}$. For $H \in M(Q)$,

$$T_H M(Q) = \{[H,\Omega] : \Omega \in \mathbb{R}^{n\times n},\ \Omega = -\Omega^\intercal\}$$

[6, Lemma 2.1.3] and its orthogonal complement w.r.t. $\mathsf{S}^2(\mathbb{R}^n) \coloneqq \{X \in \mathbb{R}^{n\times n} : X = X^\intercal\}$ is

$$N_H M(Q) = \{S \in \mathsf{S}^2(\mathbb{R}^n) : [H,S] = 0\}.$$

4.3 Lie Algebras and the Adjoint Action

1. A Lie algebra is called simple if it is non-abelian and doesn't have any non-zero proper ideals. A Lie algebra is called semisimple if it is a direct sum of simple Lie algebras.

2. For $g, h \in G$, $\mathrm{Ad}_g(h) \coloneqq ghg^{-1}$. For $x, y \in \mathfrak{g}$,

$$\mathrm{Ad}_g(x) \coloneqq \frac{d}{dt}|_{t=0}(g\exp(tx)g^{-1})$$

and $\mathrm{Ad}_g[x,y] = []$

4.4 Cartan Subalgebras

1. Let $\mathfrak{g}$ be a Lie algebra. Say $\mathfrak{g}$ is nilpotent if $\mathfrak{g}_n = 0$ for some $n \in \mathbb{N}$, where $\mathfrak{g}_0 \coloneqq \mathfrak{g}$ and $\mathfrak{g}_n = [\mathfrak{g}, \mathfrak{g}_{n-1}]$. (Equivalently, $\mathrm{ad}_{X_1} \circ \cdots \circ \mathrm{ad}_{X_n}(Y) = 0$ for any $X_1, \ldots, X_n, Y \in \mathfrak{g}$.)

2. (En)

4.5 Killing Form

1. Killing form. Let $\mathfrak{g}$ be a finite-dim Lie algebra over field K with characteristic 0. For $x, y \in \mathfrak{g}$, define the adjoint endomorphism $\mathrm{ad}_x(y) = [x, y]$. The Killing form on $\mathfrak{g}$ is $B(x, y) \coloneqq \mathrm{tr}\left(\mathrm{ad}_x \circ \mathrm{ad}_y\right)$.
   (a) $B(\cdot, \cdot)$ is bilinear and symmetric.
   (b) $B([x, y], z) = B(x, [y, z])$. (Hint: $\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ is a Lie algebra homomorphism. [5, Proposition 3.8])
   (c) If $\phi : \mathfrak{g} \to \mathfrak{g}$ is a Lie algebra automorphism, then $B(\phi(x), \phi(y)) = B(x, y)$. (Hint: $\mathrm{ad}_{\phi(x)} = \phi \circ \mathrm{ad}_x \circ \phi^{-1}$.)
   (d) $\mathfrak{g}$ is semisimple iff $B(\cdot, \cdot)$ is non-degenerate.

4.6 Homogeneous Spaces

1. Homogeneous space. Let G be a Lie group acting smoothly and transitively on a manifold M. Given $o \in M$, let $K \coloneqq \{g \in G : g \cdot o = o\}$ be the stabilizer. Then M is diffeomorphic to G/K.
   (a) Let K_o, K_t be the stabilizers of o, t. Then $K_t = gK_og^{-1}$ where $t = g \cdot o$. (Different stabilizers are conjugate to each other.)
   (b) $\Phi : M \to G/K$, $g \cdot o \mapsto gK$.
   (c) (Ex.) $G = \mathrm{SO}(n)$, $K = \mathrm{SO}(n-1)$, $G/K \cong \mathrm{S}^{n-1}$, the unit sphere in $\mathbb{R}^n$.

4.7 Symmetric Spaces

1. Symmetric space. A homogeneous space G/K is symmetric if there exists an involutive automorphism (Cartan involution) $\theta : G \to G$, $\theta^2 = \mathrm{id}$ s.t. $K = \{g \in G : \theta(g) = g\}$.
   (a) $d\theta : \mathfrak{g} \to \mathfrak{g}$, $(d\theta)^2 = \mathrm{id}$ is also an involution.
   (b) Cartan decomposition. $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, where $\mathfrak{k} = \mathrm{Lie}(K)$ and the $+1$-eigenspace of $d\theta$, $\mathfrak{p}$ is the -1-eigenspace. $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}$, $[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$. (Hint: $d\theta$ is a Lie algebra automorphism [5, Theorem 3.28], so $d\theta([X, Y]) = [d\theta(X), d\theta(Y)]$ for $X, Y \in \mathfrak{g}$.)
   (c) Let Killing form, compact/noncompact?,

4.8 Tori, Root Systems, and Weyl Groups

1. (Torus) A torus T in a compact Lie group G is a compact, connected, abelian Lie subgroup of G.

2. Let E be a finite-dim Euclidean vector space, with standard inner product $(\cdot, \cdot)$. Define the reflection about the hyperplane perpendicular to α as $\sigma_\alpha(v) \coloneqq v - 2\frac{(\alpha, v)}{(\alpha, \alpha)}\alpha$. A root system Φ in E is a finite set of non-zero vectors (roots) that satisfy:

(a) Φ span E.

(b) The only scalar multiples of a root $\alpha \in \Phi$ that is also in Φ are $\alpha, -\alpha$.

(c) For any $\alpha, \beta \in \Phi$, $\sigma_\alpha(\beta) \in \Phi$. ($\Phi$ is closed under reflection through the hyperplane perpendicular to α.)

(d) (Integrality) For any $\alpha, \beta \in \Phi$, $2\frac{(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z}$.

3. Given a root system Φ, we can always choose (in many ways) $\Phi^+ \subseteq \Phi$ (called positive roots) s.t.

   (a) For each $\alpha \in \Phi$, exactly one of $\alpha, -\alpha$ is in Φ^+.

   (b) For two distinct $\alpha, \beta \in \Phi^+$, if $\alpha + \beta \in \Phi$, then $\alpha + \beta \in \Phi^+$.

   After choosing Φ^+, $-\Phi^+$ is called negative roots.

4. Simple roots $\Delta \subseteq \Phi^+$. Each element of Δ cannot be written as the sum of two elements of Φ^+. Properties of Δ:

   (a) For every $\alpha \in \Phi$, we can write $\alpha = \sum_i c_i v_i$ with $v_i \in \Delta$, $c_i \in \mathbb{Z}$.

   (b) The coefficients c_i are either all non-negative or all non-positive.

5. The Weyl group of Φ is the group of transformations of E generated by all the reflections $\{\sigma_\alpha : \alpha \in \Phi\}$. It is a subgroup of the isometry group on Φ.

   The hyperplanes intersect E into disconnected components. Each component is called a Weyl chamber.

6. Let K be a connected compact Lie group and let T be a maximal torus in K. Define the normalizer and centralizer

$$N(T) \coloneqq \{x \in K : xtx^{-1} \in T, \forall t \in T\},$$
$$Z(T) \coloneqq \{x \in K : xtx^{-1} = t, \forall t \in T\}.$$

   The Weyl group of K (relative to T) is $W = N(T)/Z(T)$.

5 Optimization

5.1 Lower Semicontinuity and Subdifferentials

Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper lower-semicontinuous function.

1. Define $\mathrm{dom} f \coloneqq \{x \in \mathbb{R}^n : \ f(x) < +\infty\}$.

2. For $x \in \mathrm{dom} f$, the Frechet subdifferential of f at x is

$$\hat{\partial} f(x) \coloneqq \{g \in \mathbb{R}^n : \ \lim_{y \neq x, y \to x} \inf \frac{1}{\|x - y\|}(f(y) - f(x) - < g, y - x >\}$$

6 Quantum Information

6.1 Pure and Mixed States

1. A density matrix (quantum state) on a finite-dimensional Hilbert space $\mathcal{H}$ is a matrix $\rho : \mathcal{H} \to \mathcal{H}$, $\rho \succeq 0$, $\mathrm{tr}(\rho) = 1$.

2. A state ρ is called pure if $\rho = |\psi\rangle\langle\psi|$ for some unit vector $|\psi\rangle \in \mathcal{H}$. Equivalently, $\mathrm{rank}(\rho) = 1$.

3. A state is called mixed if it is not pure. Equivalently, $\mathrm{rank}(\rho) > 1$. Every mixed state can be written as a convex combination of pure states:

$$\rho = \sum_{i=1}^{r} p_i |\psi_i\rangle\langle\psi_i|, \qquad p_i \geq 0, \qquad \sum_{i=1}^{r} p_i = 1.$$

   Such a decomposition is generally not unique. One choice can be given by the spectral theorem

$$\rho = \sum_{i=1}^{\mathrm{rank}(\rho)} \lambda_i |u_i\rangle\langle u_i|,$$

   where $\lambda_i > 0$, $\sum_i \lambda_i = 1$, and $\{u_i\} \subseteq \mathcal{H}$ are orthonormal.

6.2 Separable and Entangled States

1. Suppose the Hilbert space has a tensor-product decomposition $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. Entanglement is defined relative to this choice of subsystems.

2. A state ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$ is called separable if it can be written as

$$\rho = \sum_{i=1}^{r} p_i \, \rho_i^A \otimes \rho_i^B, \qquad p_i \geq 0, \qquad \sum_{i=1}^{r} p_i = 1,$$

   where ρ_i^A and ρ_i^B are density matrices on $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively. If no such decomposition exists, ρ is called entangled. [2]

3. For a pure state $\rho = |\psi\rangle\langle\psi|$, separability is equivalent to the existence of $|\psi_A\rangle \in \mathcal{H}_A$ and $|\psi_B\rangle \in \mathcal{H}_B$ such that $|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$.

4. (Schmidt decomposition.) Every pure state $|\psi\rangle \in \mathcal{H}$ can be written as

$$|\psi\rangle = \sum_{i=1}^{r} \sigma_i |u_i\rangle \otimes |v_i\rangle, \qquad \sigma_i > 0, \qquad \sum_{i=1}^{r} \sigma_i^2 = 1,$$

   where $\{u_i\}$ and $\{v_i\}$ are orthonormal sets of $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively. The number r is called the Schmidt rank of $|\psi\rangle$.

$$|\psi\rangle \text{ is separable} \iff r = 1.$$

[2] Here $\mathrm{rank}(\rho_i^A \otimes \rho_i^B) = \mathrm{rank}(\rho_i^A)\,\mathrm{rank}(\rho_i^B) \geq 1$, so $r \geq \mathrm{rank}(\rho)$.

5. (Ex.) Pure/mixed and separable/entangled are independent concepts. Consider $\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^2$, $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$.

	Separable	**Entangled**
Pure	$\lvert\psi\rangle = \lvert 00\rangle$ also called product state	$\lvert\Phi^+\rangle = \dfrac{\lvert 00\rangle + \lvert 11\rangle}{\sqrt{2}}$ The Bell state
Mixed	$\rho = \dfrac{1}{2}\lvert 00\rangle\langle 00\rvert + \dfrac{1}{2}\lvert 11\rangle\langle 11\rvert$ convex combination of product states	$\rho_W = \dfrac{1}{2}\lvert\Phi^+\rangle\langle\Phi^+\rvert + \dfrac{1}{2}\dfrac{I_4}{4}$ mixed entangled state

Here

$$\begin{aligned}\frac{I_4}{4} &= \frac{I_2}{2} \otimes \frac{I_2}{2} \\ &= \big(\frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|)\big)\big(\frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|)\big) \\ &= \frac{1}{4}\sum_{(i,j)\in\{0,1\}^2} |ij\rangle\langle ij|\end{aligned}$$

is separable and mixed.

6.3 Partial Trace

For a bipartite system $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$,

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